

535510 Reinforcement Learning

Pre-Lecture Assignment 3

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Problem: DPG, DDPG, and Off-Policy Learning

In Lecture 11, we briefly describe the motivation for learning deterministic policies. In Lecture 12, we will discuss in detail how to learn a deterministic policy by DPG, and this involves a useful RL concept called “off-policy learning.”

Let’s try to get familiar with these concepts and get prepared for Lecture 12. [You can refer to the \(1\) slides of Lectures 11-12 and \(2\) DPG paper \(<https://proceedings.mlr.press/v32/silver14.pdf>\) as a starting point when answering the questions below.](#)

1) Deterministic PG: Please take a look at DPG expression in the slides. Can you describe each term and the intuition of behind the expression? What’s the difference between DPG and (P3) of Stochastic PG?

$$\nabla_{\theta} V^{\pi_{\theta}}(\mu) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}} \left[\nabla_{\theta} \pi_{\theta}(s) \nabla_a Q^{\pi_{\theta}}(s, a) |_{a=\pi_{\theta}(s)} \right]$$

2) Off-policy learning: Can you describe the concept of ***off-policy learning*** in 1-2 sentences? Moreover, can you provide a specific real-world example of off-policy learning? (Please explain in detail what makes it off-policy)

3) Use cases of DPG: Can you guess why DPG is suitable for solving RL problems with continuous actions?

Submission Guidelines and Remarks

- Please write all your responses within **1 page (A4 paper)**, either hand-written or typeset). The responses that go beyond 1 page will *not* be graded.
- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.
- Please submit it by **9pm, 4/14 (Tuesday)**.

